

Federated Audit Completeness: Zero-Knowledge Multi-Party Auditing via Situs Homomorphic Composition

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Abstract

The **SCX** Audit Sword establishes that any use of the Cercis Score can be independently audited — provided the auditor possesses complete data access. In federated scenarios where K parties hold private datasets and interact only through aggregated statistics, the Audit Sword encounters a structural obstacle: how does one audit what one cannot see? We formalize this as the *Federated Audit Problem* and prove three results. (i) When the Situs encoding admits a homomorphic composition operator $\oplus$ such that $\text{Situs}(\cup_k \mathcal{D}_k) \cong \oplus_k \text{Situs}(\mathcal{D}_k)$, and each party’s Cercis Score submission is verified by a zero-knowledge proof π_{ZK} , the federated audit achieves ε -equivalence to the full-data audit, with $\varepsilon \rightarrow 0$ under vanishing differential privacy budget and growing sample size. The Situs homomorphic composition construction is **conditionally rigorous** — given the operator, the ε -bound is strict; the operator’s existence is an **open problem**. (ii) Without ZK verification, the information loss is lower-bounded by $\text{const} \cdot \eta_k \cdot |\mathcal{D}_k|/|\mathcal{D}_{\text{total}}|$, a **rigorous** consequence of Theorem 3 and the data processing inequality. (iii) We outline the “Telegraph Public Good” extension where decentralized storage-layer commitments replace ZK proofs with Proofs of Storage, further tightening the ε bound — an **open problem** with a clear theoretical path. Collectively, the results establish that federated auditing is possible without raw data access, but only when the information architecture provides verifiable audit fingerprints.

1 Introduction

The SCX framework [?] established the **Audit Sword** principle: every use of the Cercis Score must be independently verifiable by any third party. This is not an optional feature — it is a structural requirement that follows from Theorem 3 (the Noise–Difficulty Unidentifiability Theorem). If you cannot audit the detector, you cannot distinguish whether it is flagging genuine label noise or merely reclassifying difficult-but-correct samples according to its own inductive biases.

The Audit Sword, however, carries a hidden premise: that the auditor has **complete access to all data**. In an increasingly federated world — hospitals that cannot share patient records across jurisdictions, financial institutions barred from exposing transaction logs, edge devices with bandwidth and privacy constraints — this premise fails. The very parties whose models most urgently require auditing (because they operate on sensitive, high-stakes data) are precisely those who cannot grant the access that auditing presupposes.

This creates the **Federated Audit Paradox**. Federated learning has developed an extensive privacy-preserving toolbox — Secure Aggregation [?], Differential Privacy [?], Multi-Party Computation [?] — but each of these tools is designed to *prevent information leakage*. The SCX requirement is the inverse: we need to *enable auditability* without compromising privacy.

Contributions.

- (1) **Federated Audit ε -Equivalence** (Theorem ??): Under Situs homomorphic composition and ZK verification, $|\text{Power}(T_{\text{fed}}) - \text{Power}(T_{\text{full}})| \leq \varepsilon(K, \delta)$, with $\varepsilon \rightarrow 0$ as $n \rightarrow \infty$ and $\delta \rightarrow 0$. [Conditionally Rigorous] / [Open] (Situs homomorphism)
- (2) **Information Lower Bound** (Theorem ??): Without ZK verification, audit power loss is bounded below by a fraction proportional to the unverified noise share. [Rigorous]
- (3) **Telegraph Public Good Extension** (Section ??): Decentralized storage commitments as a ZK replacement. [Open]

2 Preliminaries

2.1 SCX Audit Architecture

For a dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$, the **Cercis Score** $S : \mathcal{X} \rightarrow \mathbb{R}_{\geq 0}$ decomposes as

$$S(x) = Q(x) + \eta N(x), \quad (1)$$

where $Q(x) \in [0, 1]$ measures label quality and $N(x) \in [0, 1]$ measures novelty.

The **Situs operator** encodes a distribution into a metric-measure space:

$$\text{Situs}(P) = (\mathcal{X}, d_P, \mu_P). \quad (2)$$

Two distributions are compared via $d_{\text{Situs}}(P, Q) = \mathcal{W}_1(\text{Situs}(P), \text{Situs}(Q))$.

Theorem 3 (Noise–Difficulty Unidentifiability). From Cercis Score observations alone, no algorithm can distinguish label noise from intrinsic difficulty with power exceeding its significance level.

2.2 The Federated Audit Setting

Definition 2.1 (Federated Audit Problem). K parties each hold a private dataset $\mathcal{D}_k = \{(x_i^{(k)}, y_i^{(k)})\}_{i=1}^{n_k}$. The auditor $\mathcal{A}$ can access:

$$\mathcal{I}_{\text{Fed}} = \{S_k, \nabla S_k, H_{S,k}\}_{k=1}^K \cup \{M_{\Pi}\}, \quad (3)$$

where $S_k = \text{Cercis}(\mathcal{D}_k)$ is the Cercis Score computed locally. The auditor cannot access raw (x, y) pairs.

2.3 Situs Homomorphic Composition

Definition 2.2 (Situs Homomorphic Composition). A binary operator $\oplus$ on Situs spaces is **homomorphic** with respect to dataset union if

$$d_{\text{Situs}}(\text{Situs}(\cup_{k=1}^K \mathcal{D}_k), \oplus_{k=1}^K \text{Situs}(\mathcal{D}_k)) \leq \phi(n_1, \dots, n_K), \quad (4)$$

and $\phi(n_1, \dots, n_K) \rightarrow 0$ as $\min_k n_k \rightarrow \infty$.

2.4 Zero-Knowledge Audit Verification

Definition 2.3 (ZK Audit Proof). A zero-knowledge proof system for Cercis Score submissions is a triple $(\text{Setup}, \mathcal{P}, \mathcal{V})$ with completeness, soundness, and zero-knowledge properties. The commitment $c_k = \text{Commit}(\mathcal{D}_k; r_k)$ binds party k to its dataset without revealing it. The ZK proof π_k attests that S_k was correctly computed from $\mathcal{D}_k$.

3 Main Results

3.1 Federated Audit ε -Equivalence

Theorem 3.1 (Federated Audit Completeness). *Under Situs homomorphic composition (Definition ??) with error ϕ , ZK verification (Definition ??), Lipschitz audit regularity, and density regularity, the federated and full-information audit powers satisfy*

$$|\text{Power}(T_{\text{fed}}) - \text{Power}(T_{\text{full}})| \leq \varepsilon(K, \delta, \phi), \quad (5)$$

where

$$\varepsilon(K, \delta, \phi) = C_1 \cdot \phi + C_2 \cdot \sqrt{\frac{K \cdot \delta}{n_{\text{total}}}} + C_3 \cdot \frac{1 + \eta}{\sqrt{\min_k n_k}} + C_4 \cdot \text{negl}(\lambda), \quad (6)$$

with constants $C_1 = 2LM$, $C_2 = \sqrt{2/\pi}$, $C_3 = \sqrt{2 \ln(2K/\xi)}$, $C_4 = 1$.

Proof sketch. The proof employs an intermediate-statistic chain decomposition, separately bounding four error sources through triangle inequality:

1. **Situs composition error** ($C_1 \cdot \phi$): from Lipschitz continuity of the audit functional.
2. **DP distortion error** ($C_2 \cdot \sqrt{K\delta/n_{\text{total}}}$): from Gaussian DP trade-off functions.
3. **Finite-sample estimation error** ($C_3 \cdot (1 + \eta)/\sqrt{\min_k n_k}$): from Hoeffding concentration and union bound.
4. **Cryptographic soundness error** ($C_4 \cdot \text{negl}(\lambda)$): from the ZK proof system's soundness guarantee.

□

3.2 Information Lower Bound Without ZK

Theorem 3.2 (Information Loss Lower Bound). *Without ZK verification, the audit power loss satisfies*

$$\text{Power}(T_{\text{full}}) - \text{Power}(T_{\text{fed}}) \geq \text{const} \cdot \sum_{k=1}^K \frac{\eta_k |\mathcal{D}_k|}{|\mathcal{D}_{\text{total}}|}.$$

This follows from Theorem 3 and the data processing inequality: unverified Cercis Score submissions lose information proportional to the unverified noise share.

3.3 Telegraph Public Good Extension

Decentralized storage-layer commitments (e.g., Filecoin, Arweave) can replace ZK proofs with Proofs of Storage, further tightening the ε bound. This extension remains an open problem.

4 Conclusion

FA-Theorem establishes that federated SCX auditing is possible without raw data access, provided the information architecture supplies verifiable audit fingerprints through ZK proofs and Situs homomorphic composition. The ε -equivalence bound characterizes the trade-off among privacy, cryptographic soundness, and statistical accuracy.

References

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